

CIF Crystal Data

User manual and refinement parameter reference

Desktop build 1.1.0 with rietx 1.4.0 • 19 September 2026

Use this manual to import crystallographic information files, find a structure, compare its simulated powder pattern with experimental data, and perform Le Bail, Pawley or Rietveld refinement. The parameter reference explains what the refinement controls change. Pages 13–15 give additional worked guidance for searching, reviewing runs and exporting plots.

This guide describes the current Windows app, including recovery when an original CIF has moved. The rietx website may describe newer versions and features that this app does not expose.

Your first successful workflow

1 Import your structures Close the startup Quick search dialog if needed. Choose Import CIFs... and select the folder containing your CIF files.

2 Find and select a structure Quick search opens on startup. Reopen it from the toolbar whenever needed, set a criterion, click Search!, and select a result.

3 Compare the patterns Choose the correct wavelength and use Import experimental in the simulated PXRD panel.

4 Fit the experimental data Select a method and click Refine. Check the editable Bragg–Brentano and 2θ mm defaults against your instrument, then click Start refinement.

5 Review and preserve the result Inspect the fit, residual and diagnostics. Keep the checkpoint so you can resume later.

Find the instructions you need

Section	Page
Install the app and manage the CIF library	2
Compare simulated and experimental patterns	3
Run your first refinement	4
Read the fit and resume a checkpoint	5
Solve common problems	6
Technical reference and the three methods	7
Structure and measurement parameters	8
Peak shape and background parameters	9
Iteration controls and refinement switches	10
Atomic parameters and current build limits	11
Fit statistics and interpretation	12
Practical workflows and plot export	13–15

Install the app and manage the CIF library

Start the Windows app

Use either the Setup executable to install CIF Crystal Data, or the portable executable to launch it directly. The packaged app includes its refinement runtime; you do not need to install Python, npm or rietx separately. After installing an updated build, close the old app and launch the updated one.

Quick search opens automatically at startup. Close it to access the main toolbar and reopen it with Quick search at any time. Manual, immediately before About, opens this guide as a PDF in your default PDF viewer; no internet connection is needed to read it.

Import and maintain your structures

1 Choose Import CIFs... Select a folder. The app scans its subfolders for .cif files and reports successful imports and files that could not be parsed.

2 Review import failures Read the filename and reason in the results panel. Correct the source file or select a different CIF, then import again.

3 Use Refresh CIFs after changes This rescans the remembered folder. Unchanged files are skipped; new and changed files are processed. The app also attempts to refresh the remembered folder at startup.

Naming matters. Reimporting the same filename updates the existing record. Use distinct filenames for different structures. A CIF with several data_ blocks can create several structure records.

Clear CIFs removes imported database records after confirmation; it does not delete the original files. Reset search clears the current query and results, not the database. Keep the original CIF folder and experimental files as your own archive.

Find a compound

1 Open Quick search Click an element criterion box, then choose elements in the periodic-table picker. Add another criterion when needed.

2 Narrow the results Use number of elements, space-group number or symbol, reference, structural-study level, or cell-length ranges in Å. The ≠ control excludes the selected criterion; its clear button removes it.

3 Run Search! Select a result to open compound information, the structure viewer and simulated PXRD. Use column headings to sort the answer set. A blank result area before a search does not mean the imported library is empty.

Inspect and export the selected structure

The JSmol viewer provides crystallographic axis views, unit-cell blocks and optional coordination polyhedra. Its ball-and-stick connections and polyhedra are visual suggestions based on radii, not proof of chemical bonding. Open the viewer controls for the available actions.

Export CIF saves the selected original CIF block. The viewer and original-CIF export still need the source file to be accessible. Refinement can instead recover the imported structure from the database when that file is unavailable; the result displays a notice.

Learn more [IUCr introduction to the CIF standard](#)

Learn more [Jmol and JSmol documentation](#)

Compare simulated and experimental patterns

Set the simulated pattern

Select a structure from the search results. In its PXRD panel, choose the radiation wavelength and set FWHM, the full width at half maximum of a simulated peak in degrees of 2θ . A larger FWHM makes simulated peaks broader. This display control does not set the U, V, W, X and Y refinement parameters.

Wavelength choice	Value
Cu	1.5406 Å
Mo	0.7107 Å
Co	1.7902 Å
Ag	0.5609 Å

The blue simulated pattern and red experimental overlay share the 5–80° 2θ display range and are normalized for comparison. Matching peak positions is a useful starting check; a visual overlay is not a refinement.

Prepare a two column experimental file

Use .xy, .txt, .csv or .dat text with exactly two numeric columns: 2θ in degrees, then intensity. Spaces, tabs or commas separate the values. Use a decimal point, not a decimal comma. An optional heading such as “2theta intensity” and comment lines beginning with #, ;, ! or // are supported.

```
2theta intensity
10.000 120
10.020 127
10.040 142
```

The example shows the format only. A refinement needs at least 30 distinct angles inside the selected fit range. Files may contain up to 200,000 points and must not exceed the app’s 10 MB limit. Angles are sorted on import; duplicate angles must be resolved before refinement. A third uncertainty column is not accepted by this interface.

Import and compare

1 Click Import experimental Navigate to the data folder and select the experimental file.

2 Check the overlay The red pattern is scaled to a maximum of 100 for display. Confirm that the chosen wavelength and likely phase agree with the observed peak positions.

3 Replace or remove the overlay Import another file to replace it, or choose Remove experimental. The overlay is kept in the current chart session, not permanently in the CIF database.

4 Use Export .xy when needed This exports the simulated profile. The Header checkbox controls whether its header is included. It does not export the imported experimental scan or a refinement result.

Keep the original intensity scale for refinement whenever possible. The refinement window fits the imported numerical intensities, rather than the overlay normalized to 100. Its fit range can extend beyond the simulation panel’s 5–80° display.

Learn more [rietx pattern and refinement documentation](#)

Run your first refinement

Choose the method

Method	Use it when
Le Bail	You want to fit peak positions, profile and extracted intensities without refining atomic coordinates.
Pawley	You want a whole-pattern fit with reflection intensities included in the least-squares problem.
Rietveld	You have a credible atomic structure in the selected CIF and want to fit its calculated intensities to the measured pattern.

1 Select the CIF and open Refine In the simulated PXRD panel, choose Le Bail, Pawley or Rietveld from the dropdown, then click Refine. A separate window opens. A newly opened window receives the imported experimental pattern and selected wavelength.

2 Check the experimental file If needed, click Import experimental in this window. Confirm the filename and the displayed output folder. Use a writable working folder.

3 Check the structure Space group and cell parameters are prefilled from the selected structure. Confirm the phase and space-group setting. Cell lengths and angles must agree with that symmetry. Rietveld also reads the CIF atom sites.

4 Check geometry and radius All three methods start with Bragg–Brentano and a 250 mm goniometer radius. Both fields remain editable. Use the geometry and radius of your instrument; select Debye–Scherrer for a capillary experiment when appropriate.

5 Verify the required fields Asterisks mark required inputs. Confirm wavelength, polarization and minimum/maximum 2θ . The radius is optional for Debye–Scherrer, but an unknown radius limits modeling of capillary offsets.

6 Check Profile and refinement controls Open this section to see U, V, W, X, Y, zero shift, background terms, peak shape and iteration limits. Defaults are starting estimates, not an instrument calibration. The parameter reference explains them.

7 Start with a manageable free set Lattice and peak-width refinement are initially enabled; zero shift and atomic controls are initially disabled. Consider holding extra variables until the basic fit is stable. Leave Refine occupancies off in this build; see page 11.

8 Click Start refinement The plot comes into view and updates during fitting. Initial numerical-kernel compilation can make the first run slower. Use Stop and save to request a controlled stop.

Scope of this interface. It fits one phase with one constant X-ray wavelength. It does not expose multiphase fitting, neutron or time-of-flight data, an explicit $K\alpha_1/K\alpha_2$ doublet, or a general instrument-calibration workflow. Use suitable data and the correct emission-line model.

Learn more [rietx explanation of the three intensity models](#)

Learn more [IUCr Rietveld refinement guidelines](#)

Read the fit and resume a checkpoint

Read the live plot

Black points are observed intensities; the orange line is the calculated pattern; the brown line is background. Reflection ticks mark calculated Bragg positions. The gray difference trace is observed minus calculated intensity and has its own vertical scale. Do not compare its displayed height directly with a peak in the upper plot.

Rwp updates during fitting. Live frames can include trial solutions that are later rejected, so the curves need not improve on every update. After completion, inspect the full range, the final statistics, refined cell and warning messages. “Converged” describes a numerical stopping result, not proof of the structure.

Continue a run

1 Resume in the same window After a run, Resume refinement continues from its saved fitted state. You may first change allowed settings, such as iteration limits, background terms or selected refinement switches.

2 Resume after restarting the app Open a refinement window, click Load checkpoint and choose the .checkpoint.json file. Its experimental data, CIF and fitted state are restored. Click Resume refinement.

3 Start independently when intended Start new from CIF discards the checkpoint continuation and starts from the selected CIF with the current form settings. It does not reset every field to the app defaults.

Stop and save preserves the last completed stage, or the initial state if no stage completed. It does not promise to retain the very last displayed trial curve. A checkpoint is tied to its method and engine version; changing the method requires a new refinement. When a checkpoint is moved, resumed outputs are written beside the loaded checkpoint.

Keep these output files

File or folder	Purpose
.checkpoint.json	Reopen and resume. Contains data, CIF, settings and fitted state, including extracted reflection intensities.
.fit.csv	Full fitted profile: two_theta, observed, calculated, background and difference.
.refined.cif	Export of the fitted structural model. A Le Bail or Pawley export is not an independently refined atomic structure.
.report.txt and .rietx.json	Engine report and detailed numerical result. Read diagnostics as well as the headline status.
.result.json and .request.json	App result and original run inputs for traceability.
.history.jsonl, .events.jsonl, .runs and .log	Stage history, events, telemetry and process diagnostics; useful for investigating a failed run.

Each run uses a unique filename prefix beside the experimental file. The source pattern is not overwritten. Open output folder takes you to the saved files. Cancelled or failed runs may contain only partial outputs; keep the checkpoint and log.

Learn more [rietx refinement history](#)

Solve common problems

What you see	What to do
No compounds appear	Import CIFs, then run Quick search with at least one criterion. Read any import failures. Reset search clears results without deleting the library.
A text file will not import	Use exactly two numeric columns. Remove extra columns, instrument-specific headings or duplicate angles. Confirm decimal points and the 10 MB / 200,000-point limits.
Start refinement is unavailable	Import a pattern and wait for the engine check. Confirm required fields, especially geometry and Bragg–Brentano radius. A missing runtime requires the complete packaged app.
Original CIF is unavailable	The updated refinement window uses the structure stored in the database and shows a notice. To use edits made after import, restore or reimport the original CIF. Viewer and Export CIF still need the original source.
An older build shows ENOENT for a CIF	Close it and launch the updated executable containing missing-source recovery. A second click in the old executable does not repair its stored path.
The experimental file changed	Reimport it before starting. The app checks modification time and file size to avoid fitting a stale in-memory pattern.
Cannot write outputs	Move or copy the experimental file to a writable working folder, then import that copy. Check access to the displayed folder and any cloud-sync availability.
No convergence or a poor fit	Check phase, wavelength, cell, geometry and fit range first. Then review background and widths. Resume with a sensible change; raising iteration limits cannot correct a wrong model.
A checkpoint will not load or resume	Choose an app .checkpoint.json file, not .result.json. Use a compatible rietx engine version and the checkpoint's method. Do not alter embedded data or JSON fields manually.
A run stops with an engine error	Read the reported error and .log. Load the most recent checkpoint if one was saved. Retain the input files and settings when asking for help.

Separate data problems from model problems

Unmatched sharp peaks can indicate another phase. A broad residual can indicate background or profile mismatch. A systematic position offset suggests checking wavelength, cell and zero shift. These are starting clues, not unique diagnoses. Review the experimental conditions before freeing more parameters.

Keep a record of the original CIF, experimental file, wavelength and instrument geometry with your final checkpoint. Do not accept a refinement solely because its residual number is small.

Learn more [rietx manuals and examples](#)

Learn more [CIF Crystal Data issue tracker](#)

Technical reference and the three methods

The remaining sections describe the parameters used by this app's rietx adapter. "Required" means an input must be supplied; it does not mean the parameter is refined. "Refine" means the optimizer may change it. Entering a value while its refinement switch is off changes a fixed starting value.

Method	Reflection intensity treatment	Atomic parameters
Le Bail	Empirical integrated intensities are redistributed from the observed pattern between least-squares solves. They remain fixed within a solve.	Coordinates and displacement values do not determine the extracted intensities.
Pawley	Individual reflection intensities form an additional least-squares parameter block. Overlapping reflections can be strongly correlated.	An atomic model is not the source of fitted reflection intensities.
Rietveld	Intensities are calculated from the structure factors of the atomic model and a phase scale factor.	Species, positions, occupancies and displacement parameters affect the calculated intensities.

The calculation order

The app uses rietx staged plans. Parameters are released in groups, with earlier groups normally remaining free. Unchecked lattice, profile or zero-shift options remove their corresponding releases.

Le Bail and Pawley: background → zero shift if enabled → cell if enabled → W if peak widths are enabled → U, V, X and Y if peak widths are enabled.

Rietveld: scale and background → optional zero shift → optional cell → optional W → optional U, V, X and Y. Optional coordinate and displacement stages follow. The occupancy control has a current-build limitation described on page 11.

Le Bail repeats the whole plan up to Maximum Le Bail passes, retains the best Rwp result, and checks for a stable change in Rwp. Pawley and Rietveld perform one staged plan per Start or Resume operation. The reflection intensity block in Pawley is active within its stages without a separate checkbox.

What the selected CIF contributes

The cell and space group define reflection positions and symmetry restrictions. Rietveld additionally needs usable atom species and fractional coordinates. When the source file is missing, the selected database record supplies its stored structural data, and the checkpoint retains a provenance notice.

Do not treat a good Le Bail or Pawley fit as confirmation of atomic positions. To move to Rietveld, open a new Rietveld window for the appropriate CIF and transfer chosen cell/profile values manually. A method-specific checkpoint is not a cross-method conversion tool.

Learn more [rietx forward model and intensity treatment](#)

Learn more [Pawley 1981 original method paper](#)

Learn more [Le Bail Duroy and Fourquet 1988 original method paper](#)

Structure and measurement parameters

Parameter	Starting value and requirement	What it does
Space group	Prefilled from the CIF; required. Use a compatible Hermann–Mauguin symbol and setting.	Defines symmetry, systematic absences and allowed structural degrees of freedom. It is entered by the user, not automatically solved or refined.
a, b, c	Prefilled; required. Å. App input limits: 0.001–200.	Unit-cell lengths control d-spacings and peak positions. Refine lattice allows symmetry-independent lengths to change.
α , β , γ	Prefilled; required. Degrees. App input limits: 0.001–179.999; positive cell volume required.	Angles between cell axes. Keep them compatible with the space-group setting. For cubic symmetry, $a = b = c$ and all angles are 90° .
Wavelength λ	Required; Cu 1.5406, Mo 0.7107, Co 1.7902 or Ag 0.5609 Å. Inherited from the simulation panel when opened.	Sets Bragg positions through $2d \sin \theta = \lambda$. Fixed during these fits; an incorrect wavelength can bias the cell.
Geometry	Required; Bragg–Brentano by default for all three methods. Editable to Debye–Scherrer.	Selects the appropriate geometry-dependent diffraction model. Choose from the actual experiment, not from the best-looking fit.
Radius	mm; default 250 for all three methods, editable. Required for Bragg–Brentano; optional for Debye–Scherrer. Input limits 1–10,000.	Goniometer radius used by geometry-dependent displacement/offset corrections. This is not the capillary radius or sample diameter.
Polarization	Required; default 0.5, allowed 0–1. Fixed during fitting.	Fraction polarized perpendicular to the diffraction plane. The polarization factor is $K + (1 - K) \cos^2(2\theta)$; 0.5 represents the unpolarized case in this convention.
Minimum and maximum 2θ	Degrees; initialized from the imported scan and clipped to the allowed 0.001–174.999 interval. Minimum must be lower.	Select which observations enter the fit. At least 30 distinct points must remain. Refinement is not restricted to the 5–80° simulation display.

Input limits are software validation bounds, not recommended experimental values. θ is half of the plotted 2θ angle. A positive-volume cell must also satisfy the chosen symmetry. Zero shift and cell size can be correlated; a calibrated reference can help separate them.

Learn more [rietx peak-position theory](#)

Learn more [rietx intensity corrections and geometry](#)

Peak shape and background parameters

The app's common profile controls use Gaussian and Lorentzian component widths. With the separate sample-broadening terms at their initial values, the angular laws reduce to:

$$\Gamma_G^2 = U \tan^2 \theta + V \tan \theta + W$$

$$\Gamma_L = \frac{X}{\cos \theta} + Y \tan \theta$$

Γ_G and Γ_L are component full widths at half maximum in degrees of 2θ . These are not independent crystallite-size measurements. The simulated-panel FWHM is a separate Gaussian display setting and is not transferred as W .

Parameter	Default and allowed input	Effect on the fit
U	0; -0.05 to 1 deg ²	Controls the $\tan^2\theta$ contribution to Gaussian width squared, most influential at larger angles.
V	0; -0.5 to 0.5 deg ²	Controls the $\tan\theta$ contribution to Gaussian width squared. It can be negative, provided the total profile remains physical.
W	0.01; 0 to 1 deg ²	Constant Gaussian width-squared term. $W = 0.01$ gives a 0.1° Gaussian component when U and V are zero, before Lorentzian broadening.
X	0.001; 0 to 1 degree	Controls Lorentzian broadening proportional to $1/\cos\theta$. Its angular dependence resembles size broadening, but the app does not isolate instrument and sample contributions.
Y	0; 0 to 1 degree	Controls Lorentzian broadening proportional to $\tan\theta$. Its angular dependence resembles strain broadening; it is not itself a strain measurement.
Peak shape	TCH pseudo-Voigt by default; Voigt also available.	TCH approximates the convolution of Gaussian and Lorentzian profiles. Voigt evaluates that convolution directly. Shape is chosen, not optimized.
Zero shift	$0^\circ 2\theta$; -0.5 to +0.5°.	Adds a constant angular offset. Its value can be edited while fixed; Refine zero shift allows it to vary.
Background terms	3; integer 1–12.	Number of Chebyshev polynomial coefficients. Three terms mean a polynomial through degree two. More terms add flexibility but can absorb real broad features.

Background coefficients are refined automatically in all three methods. On a new run, the constant coefficient is seeded from the 10th percentile of imported intensities; the other starting coefficients are zero. A resumed run keeps fitted coefficients, adding or removing terms when requested.

Learn more [rietx peak profiles and width conventions](#)

Learn more [rietx theory and background models](#)

Iteration controls and refinement switches

Control	Default	Meaning
Max iterations / stage	100; integer 1–300	Limits work in each staged solve. It is not a time in seconds or the number of plot frames. The underlying solver may use several residual evaluations per iteration.
Maximum Le Bail passes	8; integer 1–30	Le Bail only. Limits repetitions of the whole plan so extracted intensities and profile parameters can settle together. The best pass is retained.
Intensity extraction cycles	10; integer 1–30	Le Bail: intensity-partitioning refreshes around a staged solve. Pawley: helps seed the initial intensity block; it is not the number of complete Pawley refinements.
Refine lattice	On	Allows independent cell parameters to vary under space-group constraints. Turning it off holds the currently entered values.
Refine peak widths	On	Releases W first, followed by U, V, X and Y. Turning it off uses the entered profile coefficients as fixed values.
Refine zero shift	Off	Allows the constant 2θ offset to vary. Use deliberately because it can correlate with the cell and other positional effects.
Refine atomic coordinates	Off; Rietveld only	Releases symmetry-allowed positional degrees of freedom after profile stages. A special-position atom may have no movable coordinates.
Refine displacement parameters	Off; Rietveld only	Releases atomic displacement parameters after the profile has settled. Fresh CIF imports in this build use isotropic Biso; see page 11.
Refine occupancies	Off; Rietveld only	Intended to vary site populations. The current adapter's stage mapping does not target rietx occupancy parameters correctly. Leave this control off in this build.

Changing settings before Resume

Resume keeps the checkpoint's fitted structure, instrument and reflection intensities while applying the form edits. Disabling a refinement switch holds that group at the values currently shown. Changing the space-group text regenerates reflection intensities; it is not a neutral continuation of the old reflection model.

Increase iteration or pass limits when a plausible fit is still improving. If the fit has the wrong peaks or background, correct that cause first. Do not free several strongly correlated groups solely to lower Rwp.

Advanced fitted parameter values appears after a result. It offers editable numerical values by rietx parameter path. These are optional starting-value overrides, not a general interface for changing refinement bounds, declaring constraints or enabling every feature in rietx.

Learn more [rietx staged refinement concepts](#)

Atomic parameters and current build limits

Parameter	Role in Rietveld	How this app uses it
Atom species	Determines the X-ray scattering factor.	Read from the CIF atom-site information. The app does not provide a species editor in the refinement window.
Fractional x, y, z	Locate atoms relative to the unit-cell axes and determine interference between their scattering contributions.	Read from the CIF. Refine atomic coordinates works through symmetry-allowed degrees of freedom, not three unrestricted coordinates for every site.
Site occupancy	Fractional population of an atomic site; modifies its contribution to a structure factor.	Read from the CIF. The current Refine occupancies switch must not be relied on. Where offered, an advanced .occ value is a fixed-value override, not a functioning occupancy-release control.
Biso	Isotropic atomic displacement parameter in $\text{\AA}^2$; attenuates scattering increasingly at larger $\sin \theta / \lambda$.	The fresh-CIF model uses Biso. If Uiso is provided, Biso = $8\pi^2$ Uiso. It may be released with Refine displacement parameters.
Anisotropic displacement tensor	Describes direction-dependent atomic displacements.	The database can retain tensors, but the current refinement CIF loader uses its isotropic import mode. Do not interpret this build's checkbox as a full anisotropic-tensor refinement workflow.
Phase scale factor	Adjusts the overall structural intensity scale.	Automatically seeded and refined with background in Rietveld. It is fixed in Le Bail/Pawley because their independently fitted intensities make an additional free scale redundant.

Interpret atomic parameters carefully

Occupancy, displacement and scale can compensate for one another. Keep chemistry and site multiplicities consistent, and release structural variables only when the data support them. A small numerical uncertainty does not remove uncertainty in the model or experiment.

Biso measures mean-square displacement in this model, not temperature itself. Static disorder, thermal motion and modeling deficiencies can all influence a fitted displacement parameter. Crystallographic site constraints can also prevent a requested coordinate from moving.

Limits of this app compared with rietx

The packaged engine has capabilities beyond the current window. This interface does not provide general multiphase fractions, atomic restraints, an explicit doublet, preferred orientation, absorption/extinction refinement, or a complete size-strain analysis setup. Do not infer those capabilities from the presence of related topics in the upstream manual.

The occupancy-switch mapping and isotropic CIF import behavior above are limitations of this build. They are included here so a user can distinguish what a parameter means from what this interface actually refines.

Learn more [rietx pinned structure schema and parameter definitions](#)

Learn more [IUCr practical guidance on structural refinement](#)

Fit statistics and interpretation

Quantity	Meaning and practical use
Observed and calculated intensity	The original imported intensity and the model prediction at each fitted 2θ point. Refinement does not use the comparison overlay normalized to 100.
Difference	Observed minus calculated intensity. A roughly featureless residual is preferable to systematic peak-shaped or broad features. Its plot uses a separate vertical scale.
Rwp	Weighted profile residual. The app displays a percentage. Lower indicates better agreement under the same data and weighting, but there is no universal pass value.
GOF	Goodness of fit, Rwp / Rexp in the engine. Its meaning depends on valid uncertainties and an appropriate model. A value near one alone does not prove correctness.
Cell uncertainty	Estimated standard uncertainty associated with the fit. “Unavailable” is not zero. Model inadequacy and instrumental systematic errors are not fully captured by a formal standard uncertainty.
Converged	The solver/continuation reached its stopping criterion and the app did not detect an error-level engine diagnostic. Still inspect the profile, chemistry and warnings.
max_iter, diverged or cancelled	A work limit was reached, a serious diagnostic was raised, or the run was stopped. Review the saved checkpoint and diagnostics before resuming.

How the weighting works

For two-column input, the estimated standard deviation is the square root of the observed intensity, with intensities below one treated as one for weighting. The weight is the inverse of that variance. This counting-statistics approximation may not describe normalized, background-subtracted, merged or smoothed data. The interface does not import measured uncertainties from a third column.

$$R_{\wp}(\%) = 100 \times \sqrt{\frac{\sum w_i (y_{obs,i} - y_{calc,i})^2}{\sum w_i y_{obs,i}^2}}$$

The sums run over the fitted observations. The full solver objective can also contain background or overlap restraints, so a live Rwp change is not identical to the change in every part of the objective.

Respond to diagnostics

HIGH_CORRELATION indicates parameters that may trade off; consider holding one with independent knowledge. BOUND_HIT means a parameter reached a bound; inspect physical plausibility. MODEL_FAR_FROM_DATA points to a serious mismatch. PAWLEY_OVERLAP_UNRESOLVED means individual intensities in an overlapping group are poorly separated. CAPILLARY_OFFSET_UNAVAILABLE identifies missing radius information for that correction model.

For a report or publication, keep the experimental conditions, chosen model, refined values, fit statistics, observed/calculated/difference profile, warnings and checkpoint together. Compare runs using the same input data and weighting before treating an Rwp reduction as an improvement.

Learn more [rietx estimation and fit statistics](#)

Learn more [IUCr reporting guidance for Rietveld studies](#)

Practical library and refinement workflows

Build a search in small steps

1 Begin with a known element Open Quick search and select an element in a criterion box. Run Search! before adding cell or reference restrictions. This gives you a starting answer set and helps reveal which later criterion removes a candidate.

2 Add only known information For an iron-containing compound, begin with Fe. If you know the structure is an oxide, add O in another criterion. Add the number of elements only when you know the complete composition; additional species in the CIF can make a restrictive search miss a useful record.

3 Recover from an empty answer set Clear the newest restriction and search again. Check whether a criterion is excluded with the $\neq$ control. Broaden cell-length ranges if your reference values are approximate. Review import failures if a known source file is still absent.

Select a result and compare its formula, cell and citation with the material you intended to study. Similar formulas can describe different structures. Keep filenames distinct when assembling a library: importing a different file under an existing filename can update that record.

Arrange the workspace for inspection

Drag panel dividers to give the results table, crystal viewer or PXRD panel more room. Panel sizes and table column widths are remembered locally. A focused divider responds to arrow keys; Enter or a double-click restores its default split. In the results table, Up and Down select rows, while Home and End move within the loaded rows.

In the fullscreen crystal viewer, choose a packed $1\times 1\times 1$, $2\times 2\times 2$ or $3\times 3\times 3$ unit-cell block to inspect repeated motifs. Enable the atom double-click polyhedron mode before selecting a center. Treat the radius-based connections as a visual aid and check the underlying structure before drawing chemical conclusions.

Make refinement changes easy to evaluate

1 Preserve a baseline Keep the first completed checkpoint and its report. Record the input pattern, method, wavelength, geometry and fit range so you can reproduce the comparison.

2 Change one related group For example, adjust the background terms while holding the other choices steady, or enable lattice refinement after checking the wavelength. Resume refinement from the fitted state and inspect both the residual and diagnostics.

3 Compare equivalent runs Compare Rwp only with the same observations and weighting. A change in fit range or intensity processing changes the comparison. Keep the earlier run when a lower residual comes with implausible values or stronger correlations.

To recover after closing the app, open a refinement window, load the saved .checkpoint.json file and review the restored settings before Resume refinement. Use Start new from CIF for an independent start; this keeps the current form settings, so check them explicitly.

Inspect and customize diffraction plots

Choose the view before opening the export editor

In the refinement plot, scroll over a region to zoom horizontally around the pointer, and drag to pan within the original angle limits. The intensity and difference scales remain fixed during these actions. Reset view restores the full pattern. Inspect both isolated and overlapping peaks before choosing a close-up for export.

Drag the stacked legend to a clear area. Legend entries controls which labels appear; hiding a legend entry does not hide its curve. Use Export Plot to open a separate editor with a snapshot of the current plot. The simulated PXRD panel also provides Export Plot for its simulated pattern and any experimental overlay.

An export window retains the snapshot captured when it opened. A running refinement can continue after that moment. For a final figure, wait for completion and open Export Plot again. Completed fits retain the full profile for detailed inspection; a live snapshot contains the preview data available at capture.

Work through the export settings

1 Set Size & file Choose width and height in inches, the file format and, for raster images, resolution. The preview stays within a fixed screen area and adapts to the selected aspect ratio; its screen size is not the saved pixel size.

2 Set Axes and Fonts Edit the title and axis labels. Choose whether to show Y-axis numbers or enclose the plot. Axis titles and tick numbers have separate font settings. If a selected font is unavailable, the system uses a fallback.

3 Set Ticks Major interval controls the angle spacing. With an interval of 10° and one label between majors, labels appear at 10, 15, 20, 25° and so on. X minor ticks and Y minor ticks add unlabelled marks; zero removes them.

4 Set Curves Choose the curve to edit before changing its color, line or circle style, thickness and circle diameter. Reflection ticks are available for the reflection series. Show curve changes the plotted data visibility; legend visibility is controlled separately.

5 Set Legend and Statistics Choose legend entries and font size, then drag the legend in the preview. Refinement snapshots also offer Statistics: enable Rp, Rwp and GoF and drag that group independently. A dash means that statistic is unavailable. Simulated patterns do not have refinement statistics.

6 Review and save The preview is fixed except for positioning the legend and statistics. Use Full pattern to restore the complete scan, or reopen Export Plot from a different zoomed view. Click Save plot... and select a destination. Confirm the Saved message and inspect the exported file.

Changes in this editor affect the exported presentation, not the refinement parameters. Keep labels large enough to read at the final figure size and position the legend and statistics away from peaks and reflection marks.

Choose export formats and preserve results

Select the output for its intended use

Format	Use and behavior
PNG	Default raster image. Useful for presentations and documents; preserves sharp lines without JPEG compression artifacts.
JPG	Raster image with lossy compression. Inspect fine points and text after saving.
TIFF	Raster image for workflows that request TIFF. Like PNG and JPG, includes resolution metadata.
SVG or PDF	Vector figure that remains sharp when scaled. DPI does not increase vector detail. Use when the destination accepts vector artwork.
CSV	Numerical snapshot data for analysis elsewhere. Styling, curve visibility and zoom do not crop or restyle the exported values.

For raster output, pixels equal inches multiplied by dpi on each axis. For example, 6 × 4 inches at 300 dpi produces 1800 × 1200 pixels. Width may be 4–24 inches, height 3–18 inches and resolution 72–1200 dpi. Raster exports must remain within 40 megapixels; reduce dimensions or resolution if the app reports this limit.

Understand the three numerical exports

Export .xy in the simulated PXRD panel writes the simulated profile as two columns, with an optional header. It does not include the experimental overlay. Use it when another program needs that simulation as an angle–intensity curve.

The refinement .fit.csv file is saved automatically beside the run outputs and contains two_theta, observed, calculated, background and difference. Keep this file with the checkpoint when preserving a completed fit.

CSV in Export Plot saves the snapshot data at its stored intensity scale, independently of the displayed zoom or styling. For a refinement snapshot, its columns are 2theta, Experimental, Calculated, Difference, Background and Bragg Refelctions. The last header is spelled that way in this build. Reflection angles form an independent list; a reflection on a row is not necessarily associated with that row's profile angle. Blank cells pad whichever list is shorter.

A simulated-pattern CSV instead contains separate angle and intensity pairs for simulation and experiment. Their sampling grids can differ, so compare each intensity with its own angle column. Only experimental points inside the simulation snapshot range are included.

Keep a complete record of the run

1 Preserve inputs and provenance Keep the original CIF, experimental text file and instrument settings. If refinement used a stored database structure because the source CIF was missing, retain the notice and identify that source in your records.

2 Keep recovery and numerical files Save the checkpoint with the report, fit CSV, refined CIF and detailed result. An image alone cannot resume the run or preserve all numerical settings. Retain logs for cancelled or failed runs.

3 Keep the final figure with its run Give the figure a filename that identifies the sample and run. Check that the figure came from the intended completed fit, that its labels include the angle convention, and that diagnostics were reviewed before sharing it.